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# # TWAP Edge Algorithm

## Function

Breaks down a large order into smaller, dynamically adjusted pieces and executes them at regular time intervals. The algorithm uses a **decay-based execution strategy** to prioritize earlier execution while adapting to market conditions.

## Detailed Explanation

The TWAP Edge Algorithm enhances the traditional TWAP by introducing **decay factors** to dynamically adjust execution. It balances execution across multiple intervals while ensuring the order is efficiently filled.

### 1. Order Splitting:

- The total time is divided into equal intervals, with each segment allocated an initial share of the total quantity:

**Segment Duration:**  $T_{\text{segment}} = \frac{T_{\text{total}}}{N_{\text{intervals}}}$

**Initial Segment Quantity:**  $Q_{\text{segment}} = \frac{Q_{\text{total}}}{N_{\text{intervals}}}$

### 2. Decay-Based Quantity Adjustment:

- The algorithm adjusts quantities for each segment based on the remaining time and order quantity

- Decay Formula:**

$$Q_{\text{decay}} = Q_{\text{remaining}} \cdot e^{-\text{decayfactor} \cdot \text{timeleft}}$$

Where:

- $Q_{\text{remaining}}$ : Remaining quantity to execute
- $\text{timeleft} = \frac{T_{\text{order}} - T_{\text{elapsed}}}{T_{\text{total}}}$ : Proportion of time left
- $\text{decayfactor}$ : Controls how aggressively the quantity decreases

### 3. Order Placement:

- Passive Orders:**
  - Limit orders placed at the best bid (for buys) or best ask (for sells)
- Aggressive Orders:**
  - Market orders are used for any remaining quantity if passive orders are unfilled within the interval

### 4. Dynamic Adjustments:

- The algorithm continuously recalculates the remaining quantity:



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## Real-Life Example

**Scenario:** A trader wants to buy **1,000,000 shares** of XYZ stock over **1 hour**, splitting the order into **6 intervals** with a decay factor of 0.05.

### 1. Inputs:

- $Q_{total} = 1,000,000$
- $T_{total} = 60 \text{ minutes}$
- $N_{intervals} = 6$
- $decayfactor = 0.05$

### 2. Outputs:

- Segment duration:  $T_{\text{segment}} = \frac{60}{6} = 10 \text{ minutes}$
- Initial segment quantity:  $Q_{\text{segment}} = \frac{1,000,000}{6} \approx 166,666.67 \text{ shares}$
- Decayed quantities for each segment: Using the decay formula for each interval  $t_i$ , where  $t_i = i \cdot T_{\text{segment}}$ :  $Q_{\text{decay},i} = Q_{\text{remaining}} \cdot e^{-0.05 \cdot \frac{60 - 10 \cdot i}{60}}$

### 3. Execution:

- At  $t_1 = 10 \text{ minutes}$ , the first order executes  $Q_{\text{decay},1}$  shares
- As time progresses, the algorithm recalculates the remaining quantities and places smaller orders, ensuring efficient and smooth execution

## Summary

The TWAP Edge Algorithm is designed for:

- Efficient Execution:** Prioritizing earlier segments with decay-based adjustments
- Price Optimization:** Leveraging limit orders first and switching to market orders if necessary
- Dynamic Adaptation:** Continuously monitoring and recalculating quantities based on market conditions

## Supported Directives

TWAP-Edge supports several directives that allow fine-tuning of the algorithm behavior for specific trading requirements:

### Time-Based Directives

Control the core TWAP execution timing parameters:

- time:** Object - Controls timing and chunking parameters

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(required)

- **max\_chunks** : Integer - Maximum number of chunks to execute (calculated from total\_execution\_time/chunk\_interval if not specified)

Example:

```
    "time": {
      "total_execution_time": 600,    // 10 minutes in seconds
      "chunk_interval": 60           // 1 minute chunks
    }
```

Decay Directives

Control how aggressively the algorithm distributes quantity over time:

- **decay**: Object - Controls decay-based execution
  - **factor** : Decimal - Decay factor that determines execution front-loading (0.1-1.0)
    - Higher values (0.8-1.0): More aggressive execution, heavily front-loaded
    - Medium values (0.4-0.7): Balanced execution
    - Lower values (0.1-0.3): More evenly distributed execution
  - **adaptive** : Boolean - Whether to adaptively adjust decay based on market conditions

Example:

```
    "decay": {
      "factor": 0.5,
      "adaptive": true
    }
```

Volume-Based Directives

Adjust execution relative to available market volume:

- **volume**: Object - Controls volume-based acceleration
  - **participation\_rate** : Decimal - Maximum percentage of market volume to participate in (0.01-1.0)
  - **acceleration\_enabled** : Boolean - Whether to accelerate execution when sufficient volume is available
  - **min\_volume\_threshold** : Decimal - Minimum volume that must be present to accelerate execution

Example:

```
    "volume": {
      "participation_rate": 0.3,
      "acceleration_enabled": true,
      "min_volume_threshold": 10000
    }
```

Aggression Directives



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- "aggressive" : Places orders at or beyond best bid/ask for faster execution
- "passive" : Places orders at best bid/ask (default behavior)
- "neutral" : Places orders at mid price

Example:

```
"aggression": {
  "level": "passive"
}
```

json

Trigger Directives

Define conditions for order execution or behavior changes:

- **trigger:** Object - Contains trigger conditions
  - **favorable\_ratio\_threshold:** Decimal - Executes immediately and aggressively when spread ratio crosses this threshold
  - **adverse\_movement:** Object - Protect against unfavorable price movements
    - **percent** : Decimal - Percent move that triggers protective action
    - **response** : String - Action to take ( "accelerate" , "abort" , "market" )
  - **conditional\_execution:** Object - Start execution only when conditions are met
    - **condition\_type** : String - Type of condition ( "price" , "time" , "spread\_ratio" )
    - **condition\_value** : Various - Value that triggers execution
    - **comparison** : String - Comparison operator ( "less\_than" , "greater\_than" , "equal\_to" )
    - **instrument\_pair** : String - For spread\_ratio conditions, the instrument pair to monitor

Example:

```
"trigger": {
  "favorable_ratio_threshold": 42.60,
  "adverse_movement": {
    "percent": 0.5,
    "response": "accelerate"
  },
  "conditional_execution": {
    "condition_type": "spread_ratio",
    "condition_value": 13.5,
    "comparison": "less_than",
    "instrument_pair": "BTC-USDT-PERP/ETH-USDT-PERP"
  }
}
```

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```
{
  "algorithm": "twap_edge",
  "instrument": "BTC-USDT",
  "side": "buy",
  "quantity": 10.5,
  "directives": {
    "time": {
      "total_execution_time": 600, // 10 minutes total execution time
      "chunk_interval": 60 // 1 minute chunks
    },
    "decay": {
      "factor": 0.5, // Balanced execution front-loading
      "adaptive": true // Adjust decay based on market conditions
    },
    "volume": {
      "participation_rate": 0.3, // Maximum 30% of market volume
      "acceleration_enabled": true // Accelerate if sufficient volume available
    },
    "aggression": {
      "level": "passive" // Place orders at best bid/ask
    },
    "trigger": {
      "favorable_ratio_threshold": 42.6, // Execute immediately at favorable ratio
      "adverse_movement": {
        "percent": 0.5, // Protect against 0.5% adverse movements
        "response": "accelerate" // Accelerate execution if market moves adversely
      },
      "conditional_execution": {
        "condition_type": "spread_ratio",
        "condition_value": 13.5, // Only execute when BTC/ETH ratio < 13.5
        "comparison": "less_than",
        "instrument_pair": "BTC-USDT-PERP/ETH-USDT-PERP"
      }
    }
  }
}
```

## Use Cases

### Time-Sensitive Execution

**Scenario:** A trader needs to execute a large order over a specific time period with predictable execution chunks.

**Approach:**

- Configure `time.total_execution_time` and `time.chunk_interval` to define precise execution schedule
- Set appropriate `decay.factor` to control execution pacing throughout the period
- Use `aggression.level` to manage market impact during each chunk



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Approach:

- Enable `volume.acceleration_enabled` to take advantage of volume spikes
- Set appropriate `volume.participation_rate` to limit market impact
- Configure `decay.adaptive` to true for dynamic adjustment based on real-time conditions

Spread Trading with Trigger Conditions

**Scenario:** A trader wants to execute a spread trade only when the spread ratio reaches a favorable level, with protection against adverse movements.

Approach:

- Set `trigger.favorable_ratio_threshold` to execute immediately at advantageous ratios
- Configure `trigger.adverse_movement` to protect against unfavorable spread changes
- Use `trigger.conditional_execution` with `spread_ratio` condition type to start execution only when conditions are optimal

Accelerated Decay-Based Execution

**Scenario:** A trader anticipates early opportunities but wants to maintain some execution throughout the full time period.

Approach:

- Set a higher `decay.factor` (0.7-0.9) to front-load execution
- Configure appropriate `time` parameters to define the overall execution window
- Enable `volume.acceleration_enabled` to accelerate further if market conditions permit

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